Murali Paila

Generative AI Engineer
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Summary

Generative AI Engineer who has shipped production Voice AI and RAG systems serving 3,000+ live customer calls across 60+ deployed agents. Experienced in building low-latency STT–LLM–TTS pipelines, retrieval-augmented generation, and prompt orchestration, with computer vision experience and a full-stack background (React, Node.js, MongoDB) for end-to-end delivery.

Skills

Generative AI: LLM Applications, RAG (LightRAG), Vector Retrieval, Prompt Engineering, Prompt Orchestration, Multilingual Conversational Agents, Conversation Summarization
Voice AI: Pipecat, Agora AI, Sarvam STT, Speech-to-Text, Text-to-Speech, Real-Time Streaming Pipelines
Computer Vision: YOLOX, ByteTrack, OpenCV
Languages: Python, JavaScript, TypeScript, C, SQL
Backend & Frontend: Node.js, Express.js, React, Streamlit, REST APIs, Role-Based Access Control
Databases: MongoDB, PostgreSQL, MySQL
Tools: Git, GitHub, Postman

Experience

Farm Vaidya AI Tech Pvt Ltd — Visakhapatnam
GenAI Engineer Intern — *Dec 2025 - Jun 2026*

- Architected and shipped a production no-code Voice AI agent platform powering **3,000+ live customer calls** and **10,000+ conversation minutes** across **60+ deployed agents**.
- Built a low-latency Voice AI inference graph on **Pipecat** combining STT, retrieval-augmented generation, LLM reasoning, TTS synthesis, and turn-level streaming.
- Designed vector-retrieval and prompt-orchestration flows for multilingual agents with **isolated per-customer knowledge contexts**.
- Developed an agriculture RAG chatbot with LightRAG and MongoDB session memory; integrated Agora AI and Sarvam STT for real-time voice workflows.
- Built the **Sevak Dashboard** (React, Node.js, MongoDB) to fetch Pipecat session logs and visualize conversations, summaries, and agents, with 4-level role-based access (master, superadmin, admin, user) supporting **300–400+ daily chatbot sessions**.

CSIR-CRRI — Delhi
AI Research Intern — *Apr 2025 – May 2025*

- Research intern at **CSIR-CRRI** (Central Road Research Institute, under CSIR, Govt. of India), building an AI-driven road safety analytics solution from dashcam video.
- Developed a real-time **Road Safety Scoring System** using computer vision (YOLOX, ByteTrack, OpenCV) with a Streamlit interface for live analysis and visualization.
- Built end-to-end vision pipelines for vehicle detection, multi-object tracking, ROI-based filtering, inter-vehicle distance estimation, and congestion analytics.
- Conducted four weeks of research collaboration with senior scientists, **IIT Delhi** experts, and industry partners (Palpi AI) to define the problem scope and refine system design and model evaluation.
- Achieved 65–70% detection accuracy; analyzed performance limits caused by small training datasets and documented optimization opportunities for future work.

Projects

Voice AI Agent Platform | *Pipecat, RAG, LLMs, STT/TTS, Multilingual Agents* — Farm Vaidya AI

- Production no-code platform for creating and deploying voice agents, powering 3,000+ live calls and 10,000+ conversation minutes across 60+ agents.
- Low-latency inference graph on Pipecat chaining STT, retrieval-augmented generation, LLM reasoning, and TTS with turn-level streaming.
- Vector-retrieval and prompt-orchestration flows for multilingual agents, with isolated knowledge contexts per customer.

Road Safety Scoring System | *YOLOX, ByteTrack, OpenCV, Streamlit* — CSIR-CRRI

- Real-time traffic safety analytics system for dashcam video; generates segment-level safety scores using YOLOX and ByteTrack detection and tracking, with ROI-based filtering for higher precision.
- Interactive Streamlit dashboard for safety scores and congestion analytics; achieved 65–70% detection accuracy.

Education

Gayatri Vidya Parishad College of Engineering (A) — Visakhapatnam
*B.Tech in Information Technology • CGPA: **8.5/10*** — *June 2022 – Present*

Achievements & Leadership

- Top 2 finalist at *DTI Ideathon 2024* for an innovative platform built on the MERN stack.
- Contributed a key solution idea during an **Infosys Springboard** project internship that improved the final product's accuracy.
- **Cybersecurity Lead**, GDG GVPCE; organized 3 cybersecurity awareness events.